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B.Tech. Degree IV Semester Examination April 2015

CS 1406 DATA COMMUNICATIONS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define channel capacity. What are the key factors affecting channel capacity?
- (b) What are the most significant transmission impairments?
- (c) Explain in brief (i) Line coding (ii) Block coding.
- (d) What is phase shift keying? Explain QPSK with a block diagram.
- (e) Explain the difference between asynchronous and synchronous transmission.
- (f) Explain redundancy in error detection.
- (g) Write a short note on DOCSIS.
- (h) Explain the concept of spread spectrum.

PART B

(4 × 15 = 60)

- II. (a) Discuss the different types of noises. (10)
- (b) We have a channel with 5kHz bandwidth. If we need to send a data of 150 kbps, what is the mean SNR required in dB? (5)

OR

- III. (a) Explain with necessary figures, the guided transmission media. (8)
- (b) Explain the various parameters to grade the performance of a communication. (7)

- IV. (a) Explain the following coding schemes with necessary examples and figures (8)
(i) AMI (ii) NRZ (iii) Manchester codes (iv) B8ZS
- (b) Explain pulse code modulation. (7)

OR

- V. (a) Explain the methods of LZW and Huffman coding with examples. (8)
- (b) Explain FSK. Discuss on the various types of FSK. (7)

- VI. (a) List and explain three different ways in which the CRC algorithm can be described. (8)
- (b) What are the types of errors in data communication and explain hamming code. (7)

OR

- VII. (a) Explain ARQ and stop and wait ARQ. (8)
- (b) Explain HDLC. (7)

- VIII. Explain TDM and FDM, with the help of necessary sketches. (15)

OR

- IX. (a) Write a short note on CDMA. (5)
- (b) What is DSL? Describe ADSL by defining its modulation technique. (10)

